

Computer vision assignment 1

Mece table

Name	Model build
Swetha	CNN
Tina	CNN
Comfort	KNN
Saraka	DNN
Fatemi	DNN
Gaurav	KNN
Rahul	SVM
Sherin	SVM

Chatgpt- prompt used

Swetha

first - Convert a Yelp photo . json file to CSV, randomly select 1000 image entries, ensure they exist, and use them to train a CNN model for image classification

Last- Using Keras Tuner, find the best CNN model for image classification, select the top-performing model, evaluate it with a confusion matrix and AUC, and save it as an HDF5 file.

Gaurav

first - Build and tune a k-NN model to classify images from the Yelp Photos dataset. The dataset consists of images and a JSON file containing labels. Use a pretrained CNN (ResNet18) for feature extraction

last - Modify the k-NN model training pipeline to randomly sample 1000 images, extract features using ResNet18, perform train/validation/test split, and tune hyperparameters. Ensure corrupted images are skipped

Tina

1. Summarize this project expectation. Start from step one, load the folder of yelp images and all the steps detailed explanation, processes applications and code. CNN Model with Confusion Matrix and AUC.
2. I need help with model interpretability for CNN

Comfort

First- I have a json file and a photo extract folder, how do I go about using it to build a KNN model. Add detailed explanations

Last- how can I make my model prediction more accurate

Saraka

First - Explain step-by-step how to create a DNN model for an image classification task?

Last - how to use class weights in a DNN model

Fatemi

First - Build a simple DNN model for image classification with a folder with photos and a JSON file containing labels and metadata. Consider 1,000 images of data for the model.

Last – Detect the noise, distribution of data and variance analysis to check pixel intensity distribution.

Sherin

First- Build and tune the below SVM Model with Confusion Matrix and AUC (can skip tuning on SVMs) models to predict the label of an image from a given image on the yelp Photos dataset

Last: focus on tuning but stick on svm

Rahul:

First - Train an SVM to classify JPEG images into categories.

Last- Which classes show the strongest and weakest AUC scores, and how can they be improved?